25900506 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCMAT200 - PERSPECTIVES OF MATHEMATICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

{Short Answer Questions}

Answer any **5** questions

Each question carries **2** marks

1. Plot the point $(2, -\frac{\pi}{3})$ in the polar coordinates. [A] / [CO1]
2. Define a cycloid [U] / [CO1]
3. Form an equation whose roots are the negatives of the roots of the equation $x^4 - 4x^3 + 6x^2 - x + 2 = 0$ [A] / [CO4]
4. Show that the equation $x^3 - x$ has 3 real roots. [U] / [CO4]
5. Find the area of the region R bounded by $y = x$ and $y = x^2$ in the first quadrant. [A] / [CO5]
6. Find the equivalent integral for $\int_0^3 \int_{y^2}^9 dxdy$ with the order of integration reversed. [A] / [CO5]
7. Using triple integral, find the average value of unit cube. [A] / [CO6]
8. Define Spherical coordinates. [K] / [CO6]

[2x5 = 10]

Part B

{Short Essay Questions}

Answer any **5** questions

Each question carries **6** marks

9. Write the polar equations in cartesian form. Then describe or identify the graphs of both. [A] / [CO1]

   (a) r= csc θ $e^{r \cos \theta}$ (b) r sin (θ + $\frac{\pi}{6}$) = 2

10. Sketch the line r $cos(\theta - \frac{2\pi}{3}) = 3$ and find the cartesian equations for it. [An] / [CO2]

11. If α, β, γ are the roots of the equation $x^3 + px^2 + qx + r = 0$, frame the equation whose roots are $\beta + \gamma - 2\alpha, \gamma + \alpha - 2\beta, \alpha + \beta - 2\gamma$. [A] / [CO4]

12. Compute the possible number of real and imaginary roots of the following equations. [A] / [CO4]

    1. $x^4 - 8x^3 + 14x^2 + 8x - 5 = 10.$
    2. $x^5 - 2x^4 - 5x^3 + 20x^2 - 26x = 12.$
    3. $x^7 - 12x^6 + 70x^5 - 244x^4 + 519x^3 - 668x^2 + 490x = 156.$

13. Find $\int_{-2}^{0} \int_{v}^{-v} 2dpdv$. [A] / [CO5]

14. Evaluate $\int_0^{2\pi} \int_0^{\cos 2\theta} r^2 \sin\theta \, dr \, d\theta.$ [U] / [CO5]

15. Evaluate the limits of integration r, θ and z for the solid enclosed by the cylinder $x^2 + y^2 = 4$, bounded above by the paraboloid $z = x^2 + y^2$ and bounded below by the $xy-$plane? [A] / [CO6]

16. Sketch the region of integration and evaluate the integral [A] / [CO7]
    $\int_0^2 \int_{x^2}^{2x} 4x + 3 \, dx \, dy.$

[6x5 = 30]

Part C

{Essay Questions}

Answer any **3** questions

Each question carries **10** marks

17. The ellipse $\frac{x^2}{9} + \frac{y^2}{25} = 1$ is shifted 3 units to the left and 2 units down. [A] / [CO2]

    (a) Find the equation of the new ellipse

    (b) Find the vertices, foci and centre of the new ellipse

    (c) Plot the new vertices, foci and centre and sketch in the ellipse

18. If α, β, γ are the roots of $2x^3 - 5x^2 + 4x - 1 = 0$, find: [A] / [CO3]

(a) $1/\alpha + 1/\beta + 1/\gamma$

(b) $1/(\alpha\beta) + 1/(\beta\gamma) + 1/(\gamma\alpha)$

(c) Verify that (a) × $(\alpha\beta\gamma)$ = (b) × $(\alpha\beta + \beta\gamma + \gamma\alpha)$

19. (i) Form a polynomial equation with real coefficients having $1, 3$ and $1 + i2$ as roots. [U] / [CO3]

(ii)Solve $x^3 - x^2 - 8x + 12 = 0$ given that it has a double root.

20. (i)Using polar integration, find the volume of the solid region bounded above by the paraboloid $z = 9 - x^2 - y^2$ and below by the unit circle in the xy-plane. [A] / [CO5]

(ii) Express the double integral $\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)\, dy\, dx$ in polar coordinates and then evaluate it.

21. Evaluate [A] / [CO6]

$(a) \int_0^{\sqrt{2}} \int_0^{3y} \int_{x^2+3y^2}^{8-x^2-y^2} (2x - y - z)\, dz dx dy.$

$(b) \int_0^{\frac{\pi}{4}} \int_0^{ln(sec\, y)} \int_{-\infty}^{\pi} \cos(x + y + z)\, dx dy dz.$

22. (a) Write the Jacobian of the transformation from Cartesian coordinate to cylindrical coordinate. [A] / [CO7]

(b) Find the volume of the ellipsoid $\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = 1$ using an appropriate substitution.

[10x3 = 30]